

	(Ansible, Puppet, Chef)	
2	Continuous Integration - Jenkins Introduction to Jenkins, Continuous Integration with Jenkins, Configure Jenkins, Jenkins Management, Scheduling build Jobs	8
	SECTION-II	
3	Containers and Virtual Development - Docker Docker Image, Docker Installation, Working with Docker Containers (What is Container, Docker Engine, Creating Containers with an Image, Working with Images), Docker Command Line Interphase, Docker Compose, Docker Hub, Docker Trusted Registry, Docker swarm, Docker attach, Docker File & Commands	7
4	Configuration Management Tools - Ansible Introduction to Ansible, Ansible Server Configuration, Infrastructure Management, SSH Connection in Ansible Master, YAML Scripts, Host Inventory (Hosts and Groups, Host Variables, Group Variables, Host and Group Specific Data), Ad-hoc Commands, Playbooks (Variables, Conditionals, Loops, Blocks, Handlers, Templates), Modules (Core Modules, Extra Modules), Ansible Roles	8

Practical Content:

List of programs specified by the subject teacher based on above mentioned topics.

Text Books:

1	Docker in practice by Ian Miell, Aidan Hobson Sayers, Manning Publications; 2nd edition (10 February 2019).
2	Ansible for Devops by Jeff Geerling, Midwestern Mac, LLC; 1st edition (10 October 2015).
3	Cloud Native Devops with Kubernetes by John Arundel, Justin Domingus, Oreilly pub 2019.
4	Jenkins Continuous Integration Cookbook by Alan Mark Berg, Ingram short title; 3rd edition (1 January 2017).

Reference Books:

1	Effective Devops by Jennifer Davis, Katherine Daniels, Oreilly publication 2016.
2	The Practice of System and Network Administration by Thomas A. Limoncelli, Strata R. Chalup, Christina J. Hogan, Pearson pub.
3	The Science of Lean Software and Devops Accelerate by Nicole Forsgren, Jez Humble, Gene Kim, Library of Congress Catalog-in-Publication

MOOC/Certification Courses:

1	https://www.mygreatlearning.com/academy/learn-for-free/courses/introduction-to-devops1
2	https://www.redhat.com/en/topics/devops
3.	https://intellipaat.com/jenkins-training-course/
4	https://kodekloud.com/courses/docker-for-the-absolute-beginner/
5.	https://in.coursera.org/projects/ansible-fundamentals

Question Paper Scheme:

University Examination Duration: 3 Hours

Note for Examiner: -

(I) Questions 1 and 4 are compulsory with no options.

(II) Internal options should be given in questions 2, 3, 5 and 6.

SECTION – I

Q.1 –8 Marks

Q.2 –11 Marks

Q.3 –11 Marks

SECTION - II

Q.4 –8 Marks

Q.5 –11 Marks

Q.6 –11 Marks

